

LESSON 4.11

APPLYING MULTIPLE LAWS OF EXPONENTS – PART 2



LESSON INTRODUCTION

MA.8.AR.1.1 Apply the laws of exponents to generate equivalent algebraic expressions, limited to integer exponents and monomial bases.

LEARNING TARGETS

- I can write an equivalent algebraic expression by applying multiple laws of exponents.

BENCHMARK COHERENCE

BUILDING ON

MA.7.NSO.1.1 Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational bases.

WORKING TOWARD

MA.912.NSO.1.2 Generate equivalent algebraic expressions using the properties of exponents.

ESSENTIAL QUESTIONS

- How do you apply the laws of exponents to generate equivalent algebraic expressions with mathematical fluency?

LESSON NARRATIVE

In this lesson, students bring together their understanding of applying multiple laws of exponents to write equivalent algebraic expressions. Students demonstrate their mathematical fluency (or continue to build toward fluency) through repeated practice with applying the laws of exponents to generate equivalent algebraic expressions in a series of collaborative activities in the lesson.

COMPONENT SUMMARY

4.11.1 WARM-UP (~5 MIN.)

Review of standard and scientific notation

4.11.3 ACTIVITY (20 MIN.)

Apply the laws of exponents to generate equivalent expressions through a scavenger hunt

4.11.2 ACTIVITY (20 MIN.)

Generate equivalent expressions to use in solving a riddle

4.11.4 WRAP-UP (~5 MIN.)

Self-assess mastery of learning targets from the unit

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- Scavenger hunt cards

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may confuse the operations applied to the exponents and coefficients. For example, when applying the power of products, they may add coefficients or multiply exponents. When applying the quotient of powers law, they may subtract the coefficients or divide the exponents.

THINGS TO CONSIDER

- Decrease the congestion of all the students walking around the room by having half of the class work on 4.11.2 Activity while the other half of the class completes 4.11.3 Activity. Another option is using a larger space for 4.11.3 Activity.
- Students may be tempted to reveal the riddle answer to other students. To avoid this, require students to show all of their work.

LITERACY AND LANGUAGE ROUTINES

Take Turns

Prompt students to take turns during 4.11.2 and 4.11.3 Activities. Ask students to explain their work by stating which exponent law they applied at each step. Then, encourage the other partner to share if they agree or disagree and explain why.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

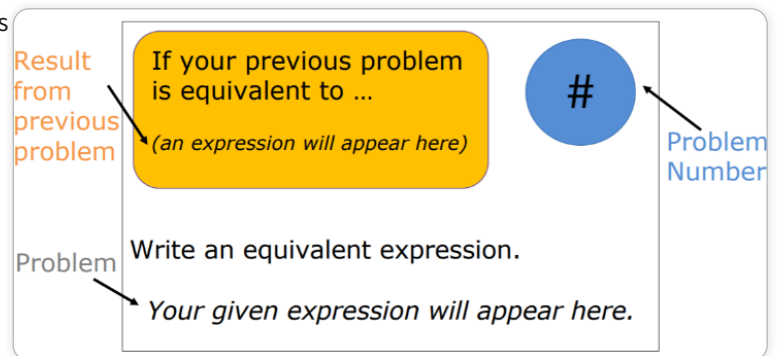
MA.K12.MTR.3.1 Mathematical Fluency

Students will exhibit mathematical fluency in applying laws of exponents to generate equivalent algebraic expressions during 4.11.2 and 4.11.3 Activities. Students have the flexibility to choose which laws they use in generating equivalent expressions and get repeated opportunities to practice efficient and generalizable methods while also reflecting with their partner on the methods they use.

DIFFERENTIATE INSTRUCTION

4.11.3 Activity provides a kinesthetic entry point, as students walk around the room to connect equivalent algebraic expressions during a scavenger hunt.

Additional differentiation opportunities are denoted throughout the lesson guide.



4.11.1 WARM-UP: READING NUMBERS

TEACHER MOVES

If time permits, consider pairing students to discuss their answers.

ENRICHMENT

Why do you think mathematicians and scientists use scientific notation instead of the standard form to express very large numbers?



4.11.1 Warm-Up: Reading Numbers

1. In the U.S., the number 42,067,513,708 is read as, "42 billion, 67 million, 513 thousand, 708."
In many Latin American countries and in the United Kingdom, the same number is read as, "42 thousand million, 67 million, 513 thousand, 708."
 - a. Explain or show how 42 billion and 42 thousand million describe the same quantity.
Answers will vary. A billion is a thousand times larger than a million, so multiplying 42 million by a thousand is equal to 42 billion.
 - b. Express the quantity 42 thousand million using scientific notation.
 4.2×10^{10}



4.11.2 Activity: Riddle Me This

- Work with your partner to generate equivalent expressions for each expression given below by applying the laws of exponents. Your results will be used to complete the riddle in question 2.

E	$a^2 \cdot (a^4)^{-2} = \frac{1}{a^6}$	N	$\frac{a^2}{a \cdot a^4} = \frac{1}{a^3}$
O	$\frac{(a^2)^4}{a^2} = a^6$	Y	$a^{-3} \cdot a \cdot (a^2)^3 = a^4$
P	$\left(\frac{a^{-2}}{a^4}\right)^3 \cdot a = \frac{1}{a^{17}}$	N	$\frac{(a^{-2})^{-4}}{a^3 \cdot a^5} = 1$
E	$\frac{a^6}{(2a^4)^3} = \frac{1}{8a^6}$	O	$(4a)(2a^{-3})^{-4} = \frac{a^{13}}{4}$
T	$-2a^{-3} \cdot (-2a^2)^2 = -8a$	X	$\frac{(5a^2)^2}{a} \cdot \left(\frac{a^6}{2}\right)^{-2} = 100a$
N	$\frac{-9a^2b^7}{3a^5b^6} = \frac{-3b}{a^3}$	T	$(a^{-4}b \cdot a^2b^{-1})^2 = \frac{1}{a^4}$
N	$(10a^3b^{-4})^2 \cdot (10a)^{-1} = \frac{10a^5}{b^8}$	P	$\frac{(8a^3b^{-2})^2}{(4a^1b^{-5})^3} = a^3b^{11}$

- Complete the answers to the riddle by writing the letters in the blanks from the expressions in question 1 that are equivalent to each expression.

In the expression x^3 , what do you call 3?

An X - P O N E N T
 $100a \quad \frac{1}{a^{17}} \quad \frac{a^{13}}{4} \quad \frac{1}{a^2} \quad \frac{1}{8a^6} \quad \frac{-3b}{a^3} \quad -8a$

In the expression y^2 , what do you call 2?

A Y - P O N E N T
 $a^4 \quad a^3b^{11} \quad a^6 \quad \frac{10a^5}{b^8} \quad \frac{1}{a^6} \quad 1 \quad \frac{1}{a^4}$

4.11.2 ACTIVITY: RIDDLE ME THIS

TEACHER MOVES

Take Turns

Prompt students to take turns as they work with their partners. Ask students to explain their work by stating which law of exponents they applied at each step. Then, encourage the other partner to share if they agree or disagree and why.

TEACHER MOVES

Monitor while students are working to determine if there are any questions that need to be addressed. This could include highlighting student voices to share rich discussions or comments overheard while they collaborated for the whole class to hear or to address common misconceptions that were evident among the majority of the class.

Decrease the congestion of all the students walking around the room by having half of the class work on 4.11.2 Activity while the other half of the class completes 4.11.3 Activity. Another option is using a larger space for the 4.11.3 Activity.

ENRICHMENT

If time allows for a debrief, students can be asked any of the following questions.

- Which law of exponents did you use at each step?
- How did you decide which order of steps to take?
- Which expression is an example of when you subtracted exponents?
- When do you subtract exponents?
- Which is an example where you added exponents?
- Did you multiply exponents on any of the expressions?

4.11.3 ACTIVITY: SCAVENGER HUNT

TEACHER MOVES

Consider giving each student “talking chips.” Every time a student talks, they can surrender a chip into a centrally located container. Tell students they must each use one talking chip for each problem. This will help ensure that all students participate in the discussions.

TEACHER MOVES

Continue to monitor while students are working to determine if there are any additional questions that need to be addressed. This could include highlighting student voices to share rich discussions or comments overheard while they collaborated for the whole class to hear or to address common misconceptions that were evident among the majority of the class.

DIFFERENTIATE INSTRUCTION

This activity provides a kinesthetic entry point, as students walk around the room to connect equivalent algebraic expressions during a scavenger hunt.

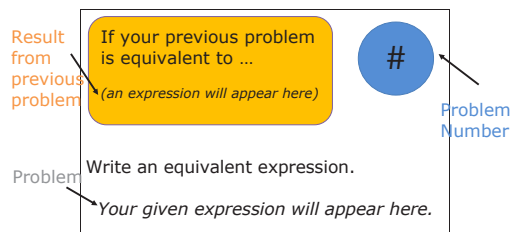
TEACHER MOVES

Students develop fluency by practicing the skills they developed during the unit. This practice will help students flexibly and accurately apply multiple laws of exponents to write equivalent algebraic expressions.



4.11.3 Activity: Scavenger Hunt

Work with your partner or group to complete the scavenger hunt. Each card in the scavenger hunt appears as shown.



Write the number of the problem you start with and the given expression in the first row of the table. Then, work with your group or partner to generate an equivalent expression using the fewest number of factors.

Look for the card with an expression equivalent to your problem to find your next problem to work on.

Repeat this process until your group has completed all the scavenger hunt problems and have ended up back at your original problem.

1 → 5 → 9 → 2 → 4 → 8 → 6 → 3 → 10 → 7 (then back to beginning)

Problem Number	Given Expression	Equivalent Expression
1		m^{-4}
5		$\frac{27m^9}{p^6}$
9		$\frac{5}{4m^3}$
2		m^6
4		$\frac{3}{4}mp^3$
8		$\frac{-p^2}{m}$
6		$\frac{4}{5}m^3$
3		$\frac{4p}{m^3}$
10		$\frac{-5m^2}{p}$
7		$27m^6p$



4.11.4 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each learning target in the table.

 I need help.  I got it.  I could help someone.

Apply one law of exponents to generate an equivalent algebraic expression.	  
Apply multiple laws of exponents to generate an equivalent algebraic expression.	  
Determine if two algebraic expressions are equivalent.	  

Answers will vary.

4.11.4 WRAP-UP: REFLECTION

TEACHER MOVES

Consider using data from this formative assessment to decide what to include in a review session before the end of unit assessment.